MARKETING CAMPAIGN PERFORMANCE & CUSTOMER ANALYTICS

Final Data Analysis Project Report

2,213	607.02	15.05%	125.4%
Customers	Average spend	Response rate	Target achievement

This project analyzes a structured customer marketing dataset covering demographics, income, household composition, customer activity, product spending, purchase behavior, campaign acceptance, complaints, and latest-campaign response. The business domain is marketing and customer analytics. The analyzed data contains 2,213 customer records and 27 source variables.

Primary question: Which customer groups respond best to the marketing campaign, and to what extent did the campaign achieve its response target?

- Evaluate the latest campaign response rate and compare it with a defined target.
- Identify high-response customer groups using spending segment, education, marital status, income, and household characteristics.
- Explore the relationship between customer income and total product spending.
- Translate the findings into practical targeting recommendations.

The dataset exceeds the project requirements of 1,000 rows and eight variables. It contains numerical and categorical fields, supports aggregation and filtering, includes measurable business metrics, and provides meaningful relationships for customer segmentation and campaign analysis.

The final cleaned workbook was checked before dashboard development. The delivered analytical table contained no blank cells, no fully duplicated rows, and no duplicated customer identifiers.

- Confirmed appropriate numeric data types for income, recency, spending, purchase counts, campaign flags, complaints, and response.
- Confirmed consistent binary coding (0/1) for campaign acceptance, complaints, and response.
- Standardized categorical labels used in the dashboard.
- Created Total_Spent as the sum of spending on wines, fruits, meat, fish, sweets, and gold products.
- Created quartile-based spending segments: Low-value (<69), Lower-mid value (69-396), Upper-mid value (397-1,047), and High-value (1,048+).
- Used distinct customer count in Power BI to prevent accidental overcounting after data transformations.
- Replaced the separate High Spender/Regular Spender slicer with the same four-level Spending Segment field used in the response chart. This removed overlapping segment definitions and made filtering consistent across the dashboard.

- The dataset does not contain campaign cost, attributed revenue, profit, advertising channel, or a control group. ROI, ROAS, cost per acquisition, and incremental lift cannot be calculated.
- The 12% response target is a management-style analytical benchmark and is not stored in the original dataset.
- Eight income observations exceed 150,000, including one value of 666,666. These records were retained but should be reviewed before predictive modeling.
- Dt_Customer represents the customer enrollment date, not the campaign date, so it should not be interpreted as a campaign-performance timeline.

Metric	Value	Metric	Value
Customers	2,213	**Total spend**	1,343,339
Average spend	607.02	**Median spend**	397
Responders	333	**Response rate**	15.05%
Complaint rate	0.90%	**Median income**	51,373

- Average spend is above median spend, showing a right-skewed spending distribution.
- Wines generated the highest recorded product spend (675,305), followed by meat products (369,488).
- Store purchases were the largest channel total (12,847), followed by web purchases (9,046) and catalog purchases (5,912).
- Response declined as recency increased: 24.1% for 0-30 days, 13.8% for 31-60 days, and 8.4% for 61-99 days.
- Customers with no children at home had average spend of about 1,104 and a 26.6% response rate.

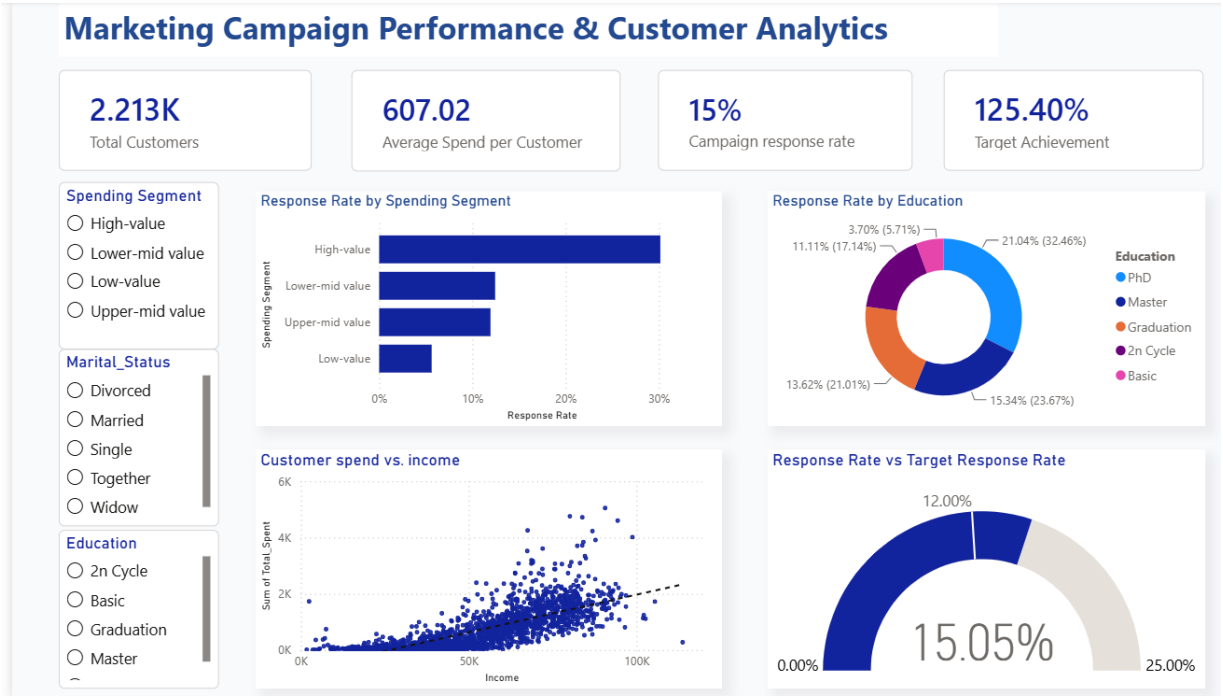


Figure 1. Final dashboard with a unified four-level Spending Segment slicer and Target Achievement KPI.

- PhD customers had the highest education-group response rate (21.0%), followed by Master (15.3%) and Graduation (13.6%). Basic had the lowest response (3.7%) and only 54 customers.
- Widow, Single, and Divorced groups had response rates above 20%, while Married and Together groups were near 11%. Smaller categories should be interpreted with sample size in mind.
- Income and total spending showed a generally positive relationship, but dispersion increased at higher income levels and extreme income values stretched the scatter-plot axis.
- Previous campaign acceptance had the strongest observed association with latest response (correlation about 0.43). Total spend was positively associated with response (about 0.26), while recency was negatively associated (about -0.20). These are associations, not causal effects.

Core insight: Recent, high-value customers with previous campaign acceptance are the strongest descriptive candidates for prioritized targeting.

The Power BI dashboard answers two connected questions: which customer groups respond best, and how much of the campaign response target was achieved. It uses KPI cards, response by spending segment, response by education, an income-versus-spend scatter plot, and a response-versus-target gauge. The slicers now use Spending Segment, Marital Status, and Education. Using the same four Spending Segment categories in both the slicer and chart eliminates the earlier overlap between Regular Spender and Upper-mid value and makes the dashboard easier to interpret.

Dashboard answer: The campaign achieved 125.4% of its response target. High-value customers were the strongest responding spending segment, PhD customers had the highest education-level response rate, and income had a generally positive relationship with spending.

The primary KPI is Target Achievement Percentage. It is calculated as the actual campaign response rate divided by the target response rate. The actual response rate is 15.05% (333 responders out of 2,213 customers), and the defined target is 12%. Therefore, the campaign achieved 125.4% of its target.

KPI formula: Target Achievement % = 15.05% / 12.00% = 125.4%

- 100% means the response target was met.
- Below 100% means the campaign did not fully achieve the target.
- Above 100% means the campaign exceeded the target.
- The result of 125.4% means the campaign performed 25.4% above the target on a relative basis.

The 12% target was selected as a clear management-style benchmark for evaluating campaign performance. It is suitable for this academic business scenario because it provides a defined expectation against which the observed response can be interpreted. The benchmark should be updated when several future campaign results or an approved management target become available.

- Campaign performance exceeded expectation: the 15.05% response rate equals 125.4% of the 12% target.
- High-value customers responded at about 30%, clearly outperforming the lower spending tiers.
- PhD customers recorded the highest education-level response rate at 21.0%, while Basic customers recorded 3.7%; group size should be considered.
- Recent customers responded more strongly: 24.1% for 0-30 days versus 8.4% for 61-99 days.
- The unified Spending Segment slicer assigns each customer to one mutually exclusive tier, improving dashboard consistency.

Recommendation	Business rationale
Prioritize recent customers	Focus first on customers whose last purchase was within 30 days.
Use value-based targeting	Prioritize the high-value spending segment and use lower-cost reactivation offers for low-value customers.
Leverage prior acceptance	Build audiences from customers who accepted earlier campaigns.
Personalize product offers	Use wine and meat affinity for relevant bundles and cross-sell recommendations.
Improve measurement	Capture campaign cost, channel, date, and a control group in future campaigns.
Monitor target achievement	Use 100% as the point at which the target is met and track overachievement above 100%.